Joshua Dinn

SOFTWARE ENGINEERING

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EDUCATION

Monash University, Melbourne, Australia — Mar 2023 – Oct 2027

Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)

WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

EXPERIENCE

Monash DeepNeuron, Optimised Computing Member — Sep 2025 – Present

- Software team member on HarvestAI, an autonomous strawberry-picking robotic arm. Building the computer-vision perception module: curating and augmenting the training image set, then training an object detection model that locates individual berries in a live camera feed and classifies ripeness in real time, with training accelerated on HPC clusters.
- Perception output feeds the grasp-planning stage as 3D coordinates the arm acts on, so detections have to hold up in the real scene rather than only on a held-out test set.
- Building the on-device AI memory system for AR glasses: a vector database with HNSW indexing and approximate nearest-neighbour search powering a retrieval-augmented generation (RAG) pipeline that grounds responses in prior context with low latency.
- Teaching Assistant for an 8-week C++ and accelerated computing workshop on GPU programming, CUDA, and HPC, run with NVIDIA, the Pawsey Supercomputing Centre, and NCI. NVIDIA CUDA certified.

Collingwood Football Club, Data, Analytics and Technology Intern — Mar 2026 – Jun 2026

- Designed and implemented a repository synchronisation workflow between the club's web and mobile codebases, removing a manual step from every deployment.
- Built a membership churn prediction model in XGBoost, engineering the feature pipelines from historical transaction and engagement data to flag at-risk members for the commercial team.
- Built an interactive dashboard visualising churn drivers and SHAP feature importance, giving stakeholders a self-serve view into retention risk.

Monash eSolutions, AI Fitness Trainer — Jul 2026 – Present

- Run in-person sessions teaching students and staff how to use AI effectively, ethically, and in everyday study and work.
- Helped deliver a university webinar on AI to an audience of 200+ attendees.

Outlier AI, AI Trainer (Mathematics) — Jun 2024 – Dec 2025

- Reviewed and corrected AI-generated mathematical solutions written in LaTeX to improve accuracy and reasoning quality.
- Wrote detailed feedback on computational and logical errors across advanced problem sets.

PROJECTS

StdOut: Real-Time AI Technical Interview Simulator — Feb 2026

- Built a real-time interview simulation platform with a 3-person hackathon team using React, Node.js, MongoDB, and WebSockets.
- Engineered a low-latency pipeline combining live transcription, AI-driven conversation, and in-browser coding, handling streaming updates asynchronously so they never blocked the session.
- Built a Python code-execution terminal with event-based tracking of code diffs and session timelines, enabling full replay of a candidate's coding process.

JackTen: No-Limit Hold'em Poker Bot — Ongoing

- Building a poker decision engine from first principles in Python: hand evaluation and equity first, then the expected value, pot odds, and minimum defence frequency calculations that drive every action.
- Modelling opponents with Bayesian range updates from observed actions, combined with tracked statistics (VPIP, PFR, 3-bet%) to move from a balanced baseline strategy to exploitative play.

Interactive Maze Solver & Algorithm Visualiser — Oct 2025

- Built a pathfinding platform visualising 7 algorithms across procedurally generated mazes.
- Added automated benchmarking and metrics tracking across 125,000+ simulated maze runs, with property-based tests (Hypothesis) covering maze generation and solver invariants.

SKILLS

Languages: Python, C++, JavaScript, TypeScript, Java, SQL, MATLAB, R

Frameworks & Tools: React, Node.js, PyTorch, OpenCV, MongoDB, AWS, Linux, Git, pytest, Hypothesis

Data: pandas, NumPy, scikit-learn, XGBoost, statsmodels

Interests: System design, machine learning, statistical modelling